Introduction

- Agriculture is becoming an increasingly important sector in the Zambian economy since the mineral sector, which was the backbone of the economy.
- The agriculture sector generates about 18% to 20% of the country's GDP and provides a livelihood for more than 60% of the population.
- It employs about two-thirds of the labor force.
- In 1991 there was a major shift in the country's agricultural policy with the liberalization of the marketing of agricultural inputs and produce.
- This new policy has resulted in low agricultural productivity as subsistence farmers, who make up about 75% of the entire farming community in Zambia, find difficulty in procuring farm necessities such as seeds and fertilizer in terms of accessibility and funds.

- Their difficulties are further compounded by the erratic seasonal rainfall patterns which have been experienced by the country in the last 20 years.
- Almost all subsistence farming in Zambia relies on rainfall.
- Over the last two decades, the frequency of extreme climate events such as high surface temperatures, floods and droughts has increased over the entire globe.
- Although such extreme events are attributed to climate variability, they also signal that the earth is going through longterm climate changes in mean temperature and rainfall norms.

Over View of Zambia

- Physical attributes Zambia covers a land area of 752,615km2 and lies between 22° to 34° east of Greenwich and 8° to 18° south of the equator.
- The country consists mostly of plateau with an elevation between 950 meters to 1,500 meters above sea level.
- The country has a sub-tropical climate and vegetation.
- There are three distinct seasons:

- A warm wet season stretching from November through April during which 95% of the annual precipitation falls
- A cool dry winter season from May to July with the mean temperature varying between 15°C and 27°C
- A hot dry season prevailing from August to October with an average maximum temperature of 27°C to 32°C.
- The annual rainfall varies from over 1200mm in the north to about 700mm in the central part of the country and less than 700mm in the south.

Agriculture in general

- The agricultural sector in Zambia supports livelihoods of 85% of the population.
- Maize is the principal cash crop (>65% of cropped land) as well as the main staple crop.
- Per capita consumption of maize is estimated at 105 kilograms annually, most of which is ground into meal and consumed as stiff porridge, or fermented for beer, with by-products used as livestock feed.
- Other important crops include soybean, cotton, sugarcane, sunflower, wheat, sorghum, pearl millet, cassava, tobacco and various vegetable and fruit crops.

Agriculture in general contd

- Agriculture is relatively unimportant in Zambia's economy compared to mining (primarily copper).
- It contributes less than 20% of Gross Domestic Product.
- But Zambia has potential for significant increases in agricultural output; currently, less than 30% of potentially arable land is cultivated.
- In the past, the agriculture sector suffered from low product prices, difficulties in availability and distribution of credit and inputs, and shortage of foreign exchange.

Types of farmers in Zambia

- There are three major categories of farmers in Zambia, defined in terms of the land area cultivated by each farmer.
- ***Small-scale farmers***, who are the vast majority, cultivate less than five ha, use few external inputs, and consume most of their produce, occasionally entering the market to sell any surplus.
- The hand hoe is the predominant means of cultivation.
- ***Medium-scale farmers*** cultivate between 5 to 20 ha.
- They use improved seeds and fertilizers and sell most of their production.
- farmers commonly use a combination of manual, animal draft power and tractors.

Types of farmers in Zambia

- ***Large-scale commercial farmers*** plant over twenty ha annually.
- These farmers apply high levels of purchased inputs and use oxen or machinery for farm operations.
- They produce almost exclusively for direct market sale or feed their grain to livestock kept on the farm.
- Large-scale farmers make up only 4% of farm households, but cultivate 22 per cent of all cropped land.

Production and yield

- Crop yields depend on the production circumstances, with higher yields on commercial farms and low yields on small-scale farms.
- Annual maize production in Zambia was on average 1.1 Mt in the period 2000 to 2010, and average yields of about 1.5 t/ha that have not significantly changed over the past 20 years.
- Cultivated maize area has been on average 700,000 ha between 1987 and 2007, with largest area of 900,000 ha in 2008.

Production and yield

- The yield gap between actual yields and potential yields remains wide.
- Zambia produced an annual average of about 24,000 metric tons of sorghum and 42,000 metric tons of millet between 1987 and 2008.
- Sorghum and millet national average yields are very low, about 0.55 and 0.65 t/ha, respectively.

Agro-ecological zones and soils

- Zambia is divided into three major agro-ecological regions (Regions I , II and III), which are primarily based on rainfall amount but also incorporate soils and other climatic characteristics.
- **Semi-arid Region I** includes areas of southern, eastern and western Zambia: Zambia's valleys at 300-800 m altitude mostly lie in Region 1.
- Mean annual rainfall in Region I ranges from 600 to 800 mm.
- The growing season is relatively short (80-120 days) and risky for crop production, as poorly distributed rains result in crops enduring frequent dry spells.
- Region I contains a variety of soil types, ranging from slightly acidic loamy and clayey soils with loam topsoil, to acidic sand soils

Agro-ecological zones and soils

- Characteristics of these soils which have significant constraints for crop production, include: erosion, limited soil depth in hilly and escarpment areas, poor physical properties that make it difficult to till especially on cracking clay soils, crusting, and low water holding capacities in sandy soils.

- **Region II** includes much of central Zambia, with most of Central, Southern, Eastern and Lusaka provinces.
- It contains the most fertile soils and most of the country's commercial farms. Annual rainfall in Region II averages 800-1000 mm, and the growing season is 100-140 days long.
- Distribution of rainfall is not as erratic as in Region I, but dry spells are common and reduce crop yields, especially on the sandier soils.
- Average mean daily temperatures range from 23- 26°C in the hottest month October to 16-20°C in the coldest months of June and July.

- The most common soils in Region II are red to brown clayey to loamy soil types that are moderately to strongly leached.
- Physical characteristics of the soils that affect crop production, include low water holding capacity, shallow rooting depth, and top soils prone to rapid deterioration and erosion.
- These soils also have low nutrient reserves and retention capacity, are acid, have low organic matter and nitrogen content, and are phosphorus-deficient.

- **Region III,** the high-rainfall area, lies in a band across northern Zambia, including the Northern Luapula Copperbelt, Northwestern provinces and some parts of the Central province.
- This region receives over 1000 mm of precipitation each year, and the growing season ranges from 120-150 days.
- Soils in Region III are highly weathered and leached, and characterized by extreme acidity.
- Consequently, the soils have few nutrients available for plant growth, and are high in exchangeable aluminum and manganese, both of which are toxic to most crops unless soils are limed to increase pH

Major cropping systems

- **Region I**
- Region I has predominantly small-scale farmers in the major valley systems. In the Luangwa Valley, sorghum, finger millet and maize are the major starchy food crops, while groundnuts, cowpeas and pumpkins are also grown.
- Farmers use hand hoes for cultivation. Goats and chickens are commonly kept by farm households, and some farmers have a few cattle.
- Other areas of the region mainly produce bulrush millet, sorghum, and cassava.

Major cropping systems

- **Region II**
- Zambia's large commercial farmers are concentrated in Region II.
- Their farming systems are mechanized and highly diverse, cultivating maize, soybeans, wheat, cotton, tobacco, coffee, vegetables, and flowers, and breeding livestock.
- Besides these large-scale systems, there are also small- and medium-scale farmers in the region.
- Maize is the main staple crop in these systems in Central and Eastern provinces.
-

Major cropping systems

- Beans, groundnuts, pumpkins, and cassava leaves are grown to diversify diets.
- Other crops include cotton, sorghum, soybeans and sunflower.
- Cattle, chickens, goats, pigs and sheep are common.
- Farmers also grow tobacco.
- Cattle are important for traction, meat, milk and manure.
- The major constraints to increase crop production in Region II are the lack of low-cost biocides to control pests and diseases, soil degradation, and the depletion of soil fertility.

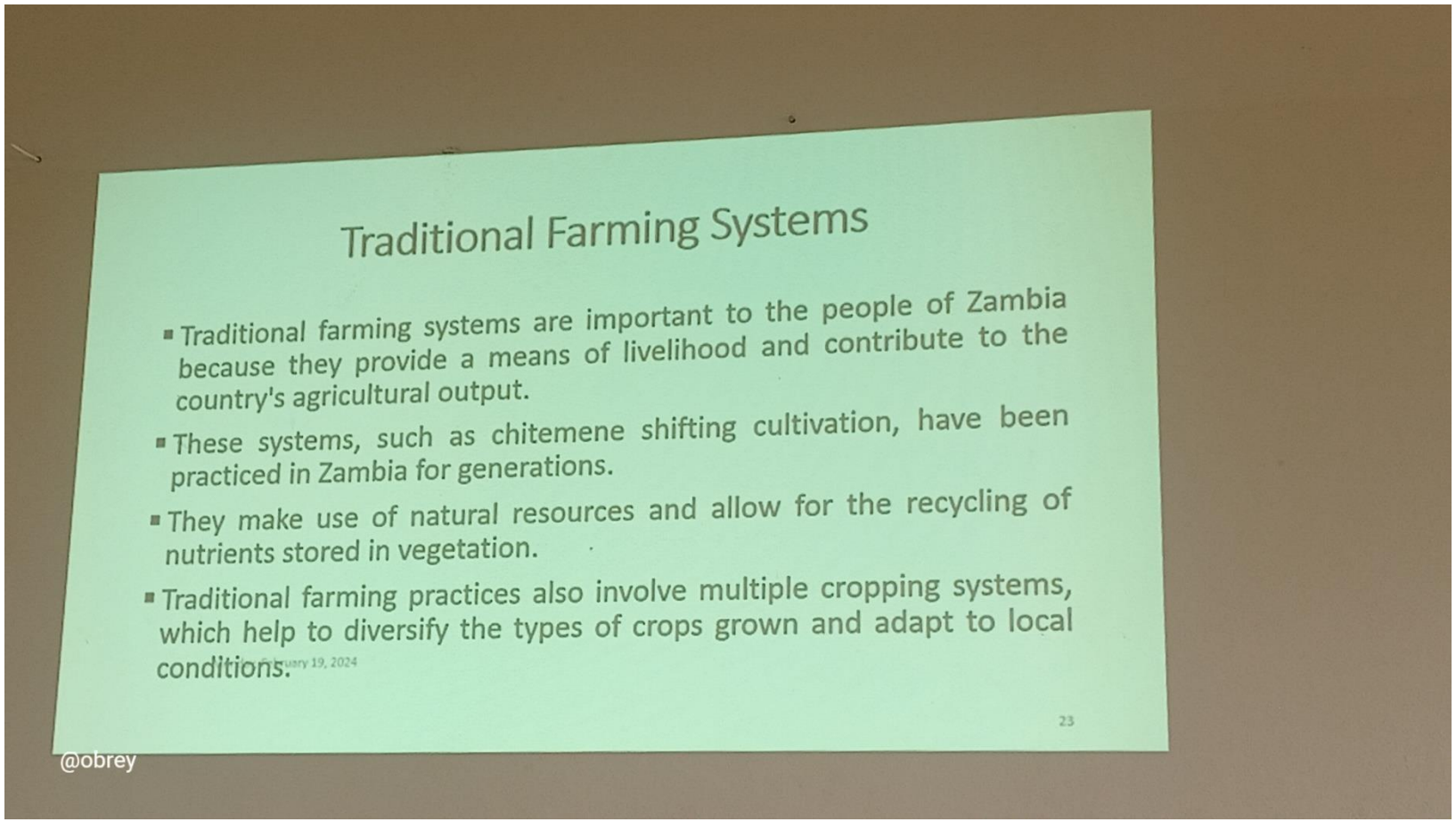
Major Cropping systems

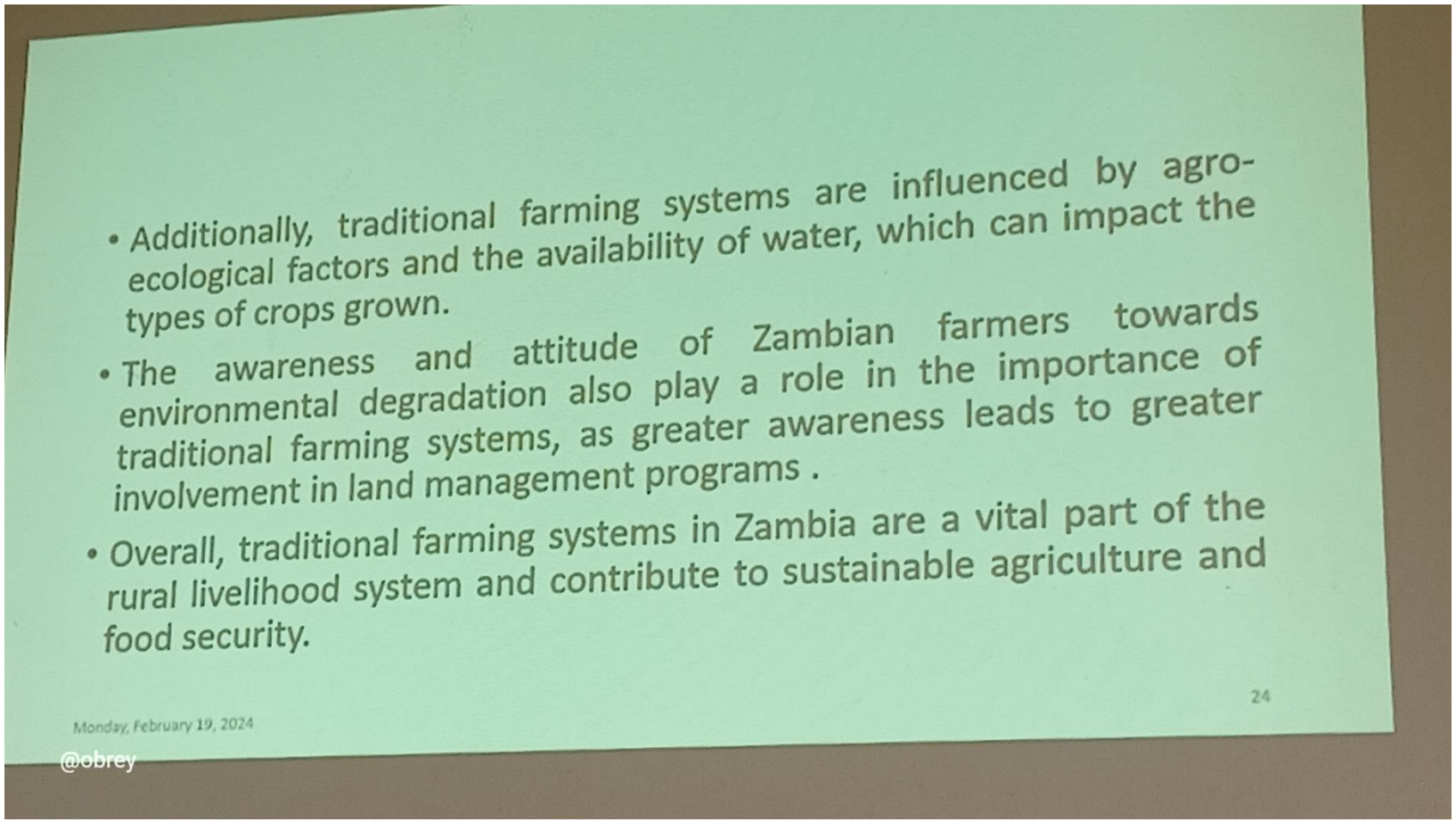
- Grass is cut and buried at the end of the rainy season and allowed to decompose. The composted material is spread before the next planting season onto frequently mounded fields.
- Grass is cut and buried at the end of the rainy season and allowed to decompose. The composted material is spread before the next planting season onto frequently mounded fields.
- Principal crops in the hand hoe system of Northern, Luapula and Northwestern provinces are cassava, landrace maize varieties, sweet potatoes, pumpkin, finger millet and beans.
- Most farmers have chickens and a few goats, but other livestock is uncommon. The existence of tsetse fly in some areas limits opportunities for cattle production.

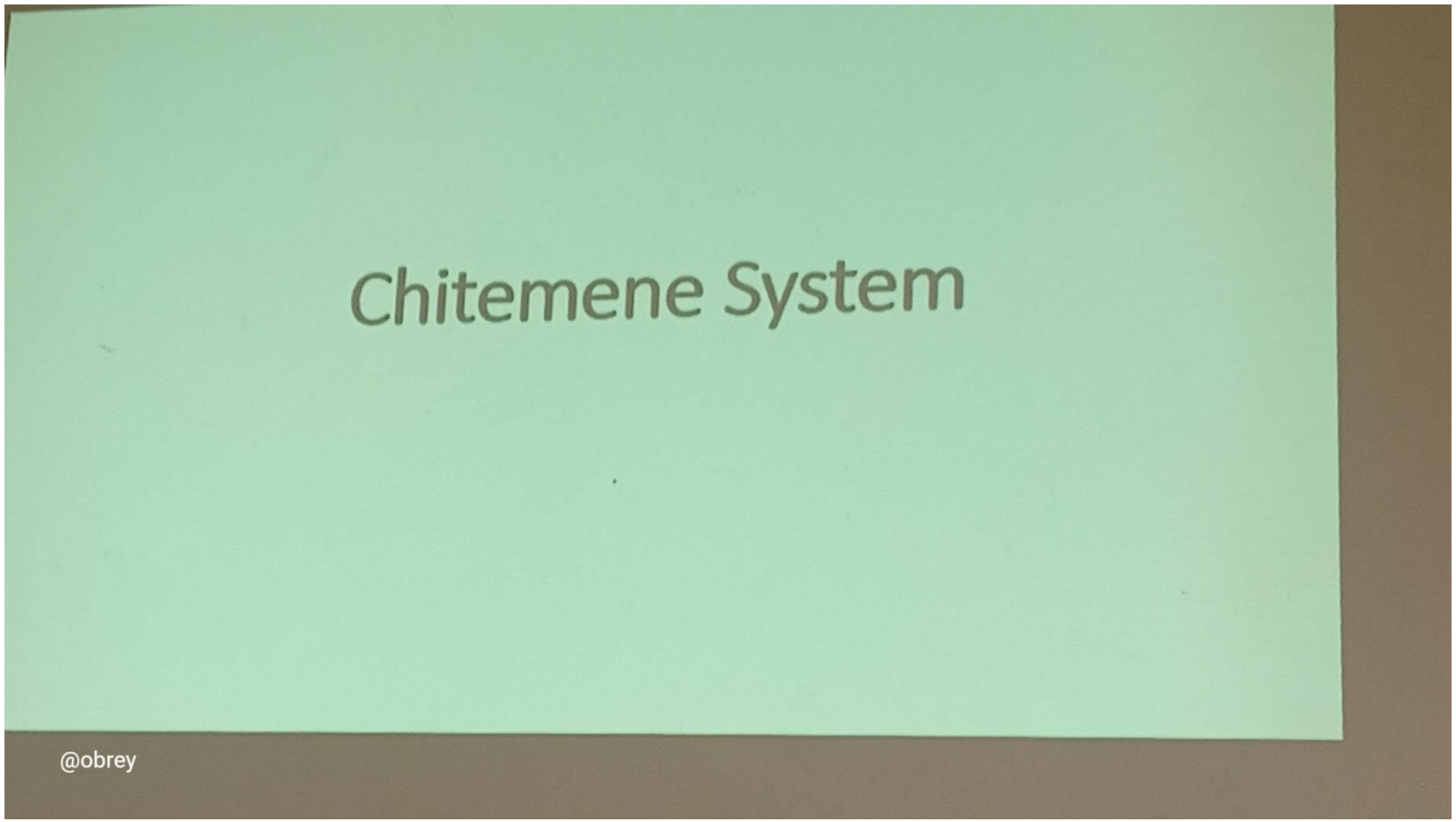
Traditional Farming Systems

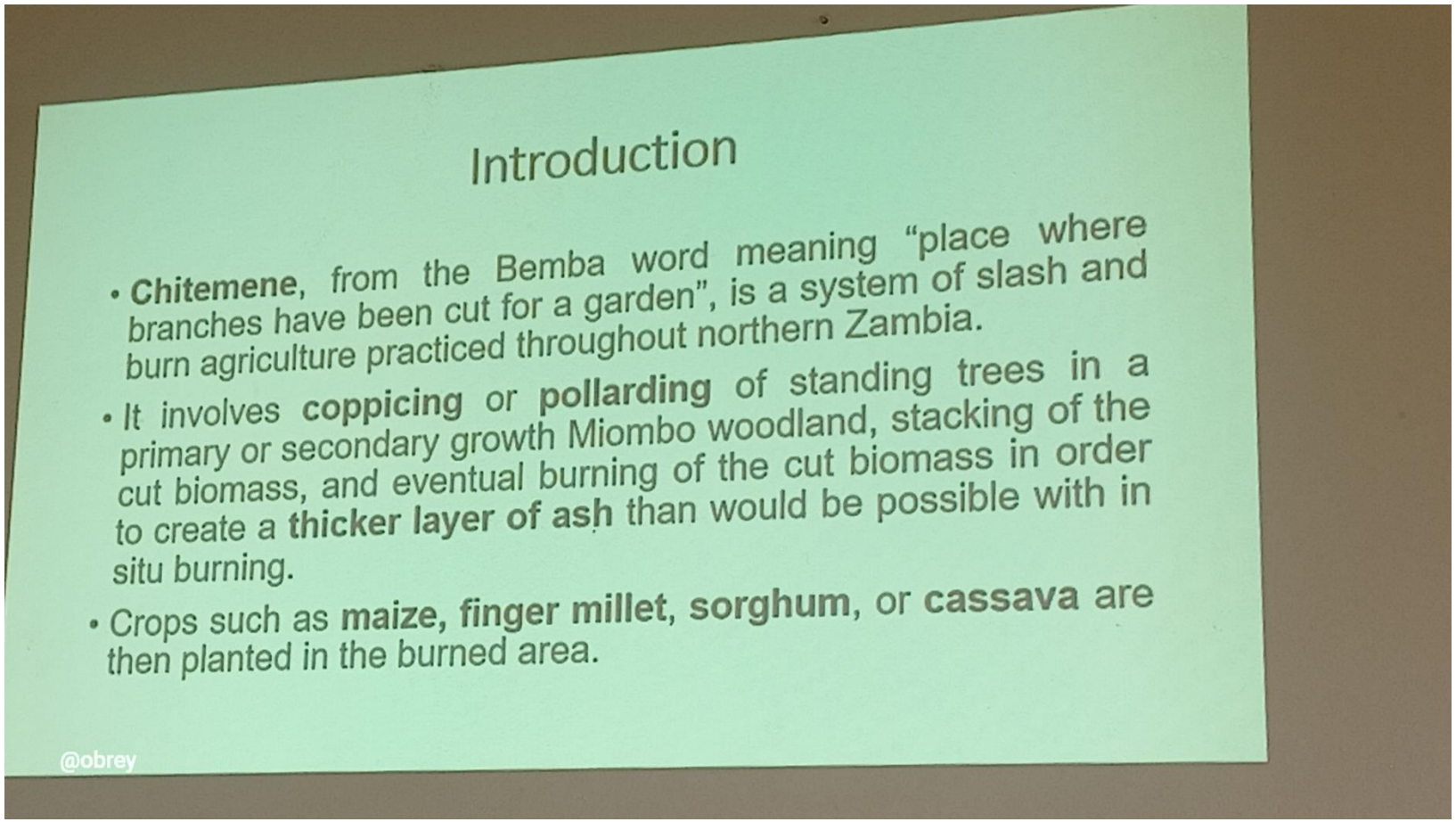
- Traditional farming systems are important to the people of Zambia because they provide a means of livelihood and contribute to the country's agricultural output.
- These systems, such as chitemene shifting cultivation, have been practiced in Zambia for generations.
- They make use of natural resources and allow for the recycling of nutrients stored in vegetation.
- Traditional farming practices also involve multiple cropping systems, which help to diversify the types of crops grown and adapt to local conditions.

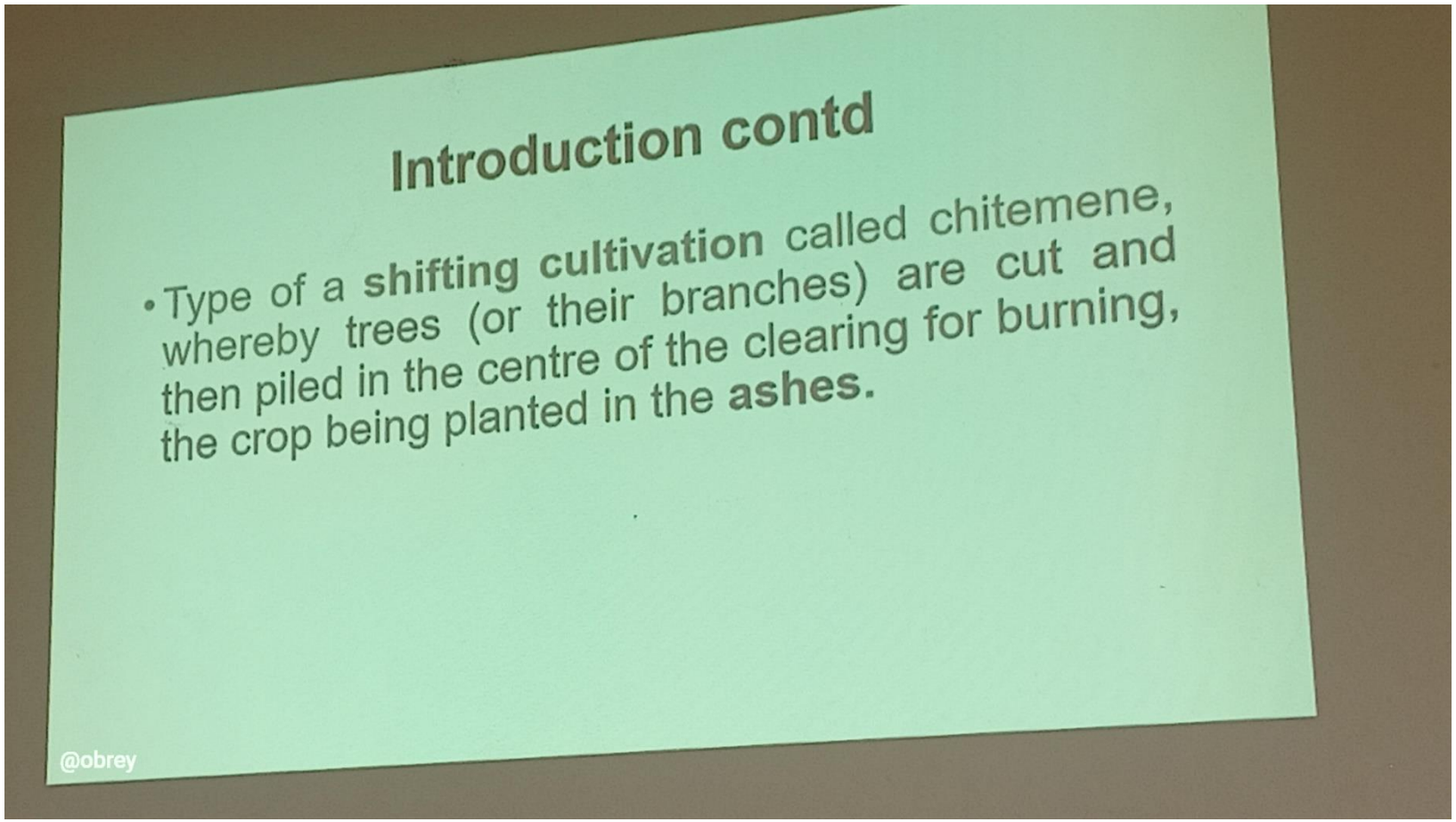
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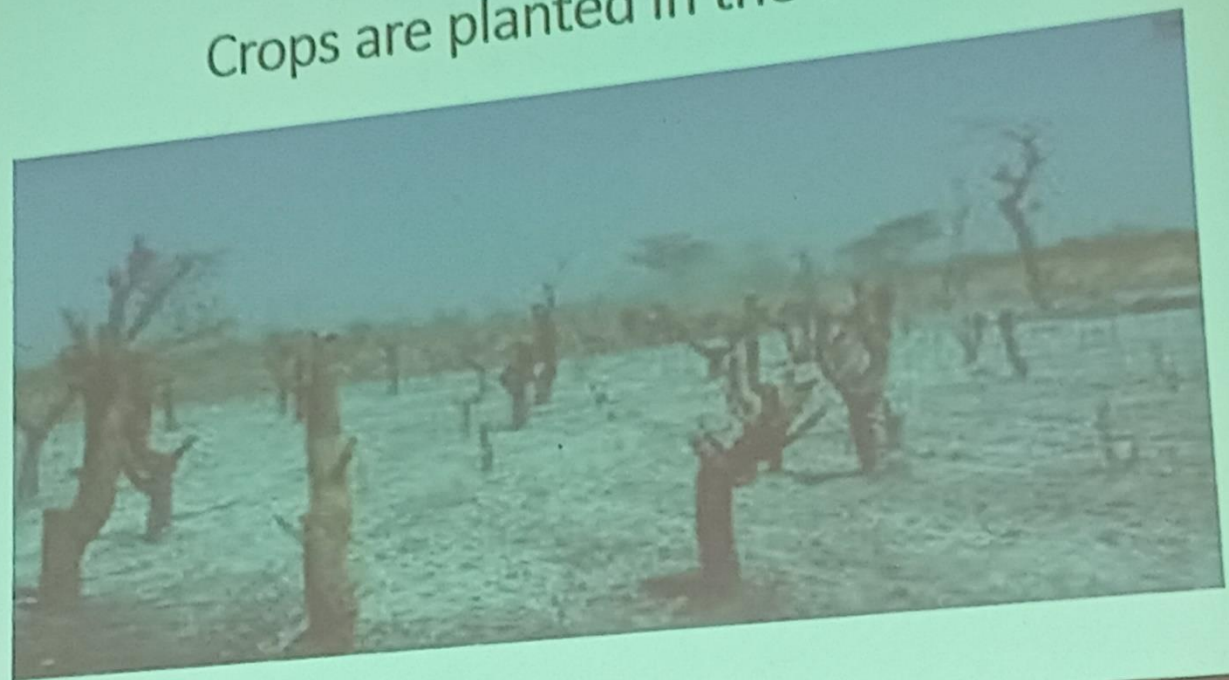

• Additionally, traditional farming systems are influenced by agro-ecological factors and the availability of water, which can impact the types of crops grown.
• The awareness and attitude of Zambian farmers towards environmental degradation also play a role in the importance of traditional farming systems, as greater awareness leads to greater involvement in land management programs .
• Overall, traditional farming systems in Zambia are a vital part of the rural livelihood system and contribute to sustainable agriculture and food security.
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Chitemene System
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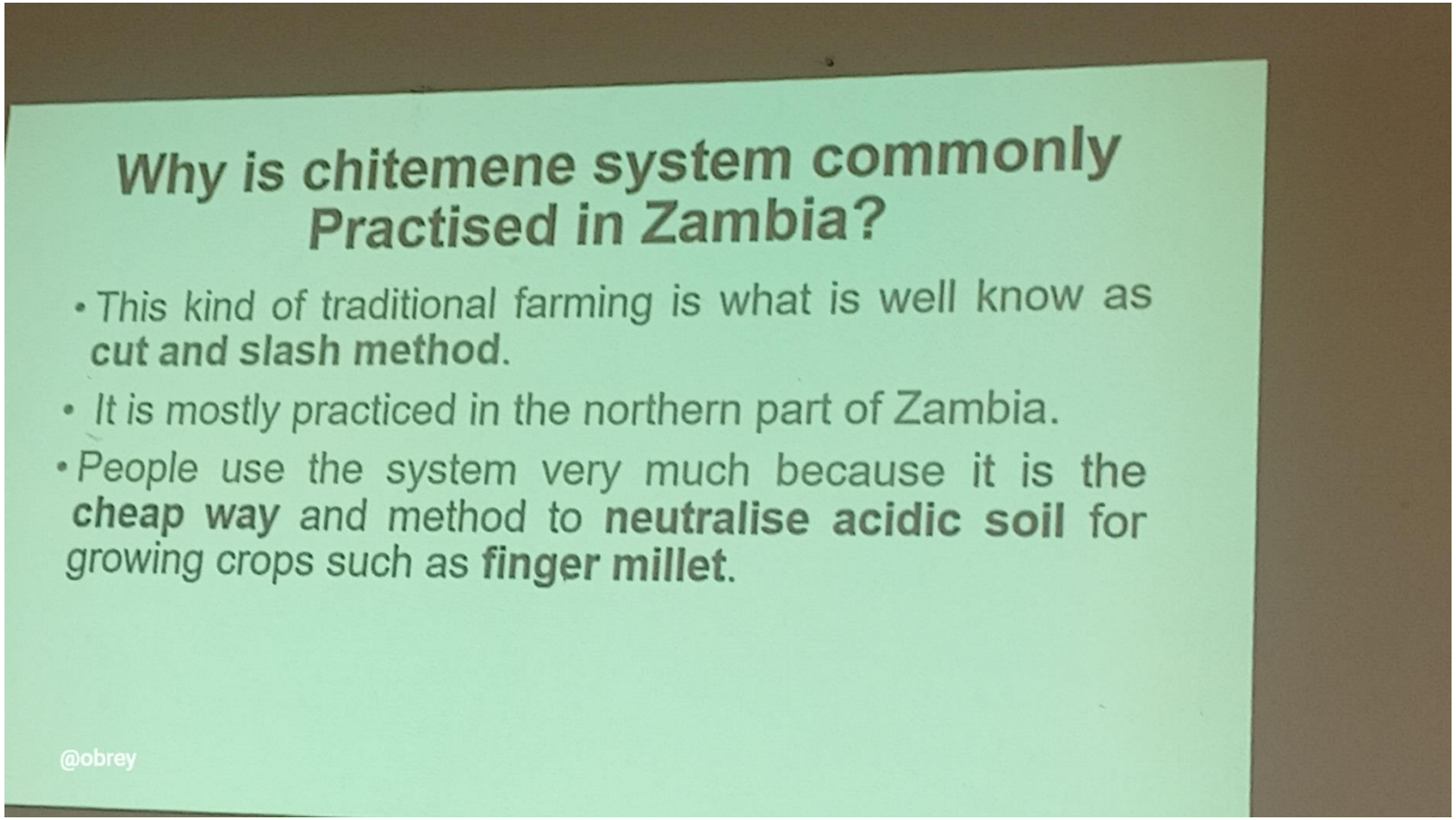

Introduction
• **Chitemene**, from the Bemba word meaning "place where branches have been cut for a garden", is a system of slash and burn agriculture practiced throughout northern Zambia.
• It involves **coppicing** or **pollarding** of standing trees in a primary or secondary growth Miombo woodland, stacking of the cut biomass, and eventual burning of the cut biomass in order to create a **thicker layer of ash** than would be possible with in situ burning.
• Crops such as **maize, finger millet, sorghum**, or **cassava** are then planted in the burned area.
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Introduction contd
• Type of a **shifting cultivation** called chitemene, whereby trees (or their branches) are cut and then piled in the centre of the clearing for burning, the crop being planted in the **ashes.**
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Crops are planted in the ash



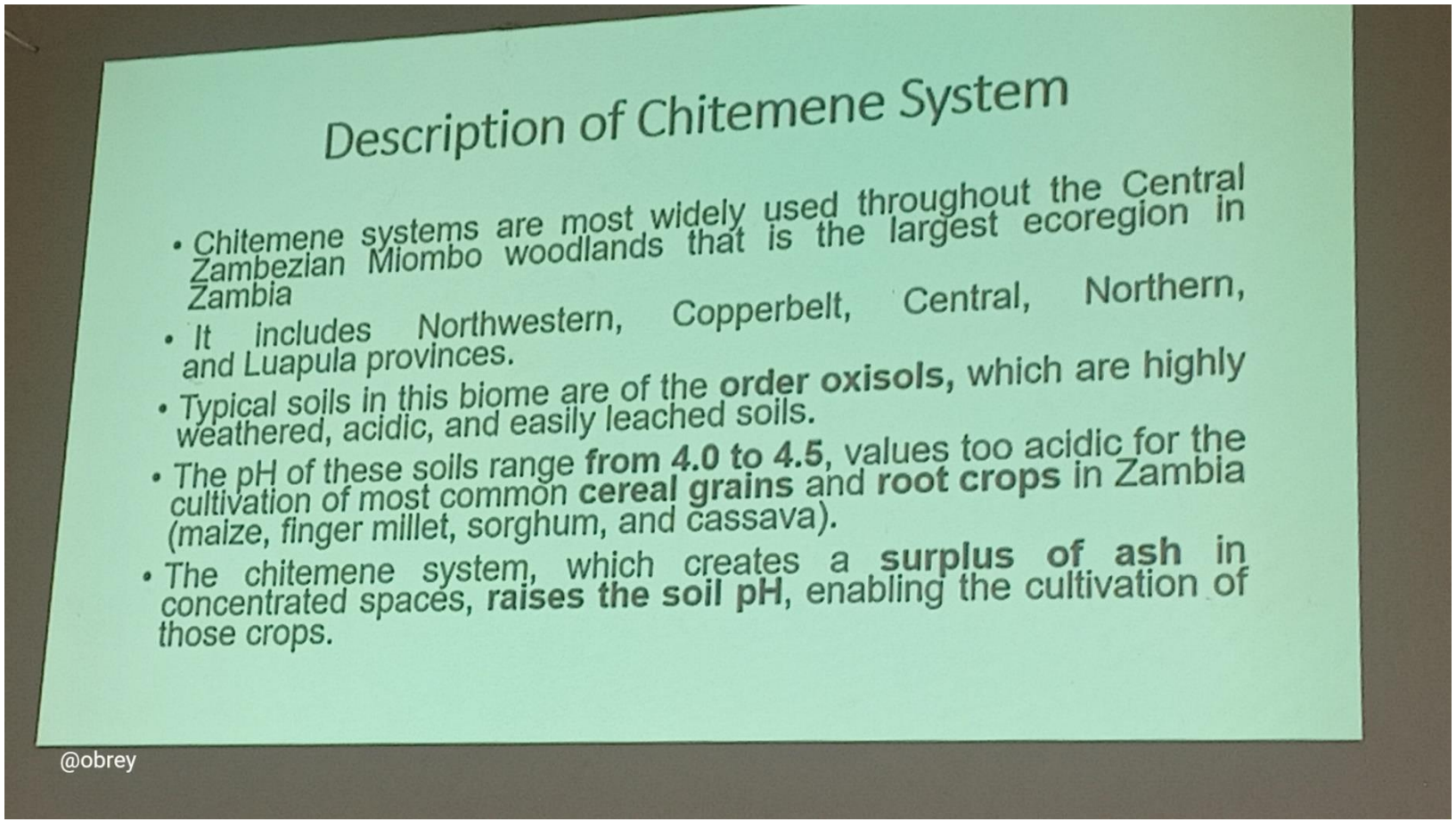
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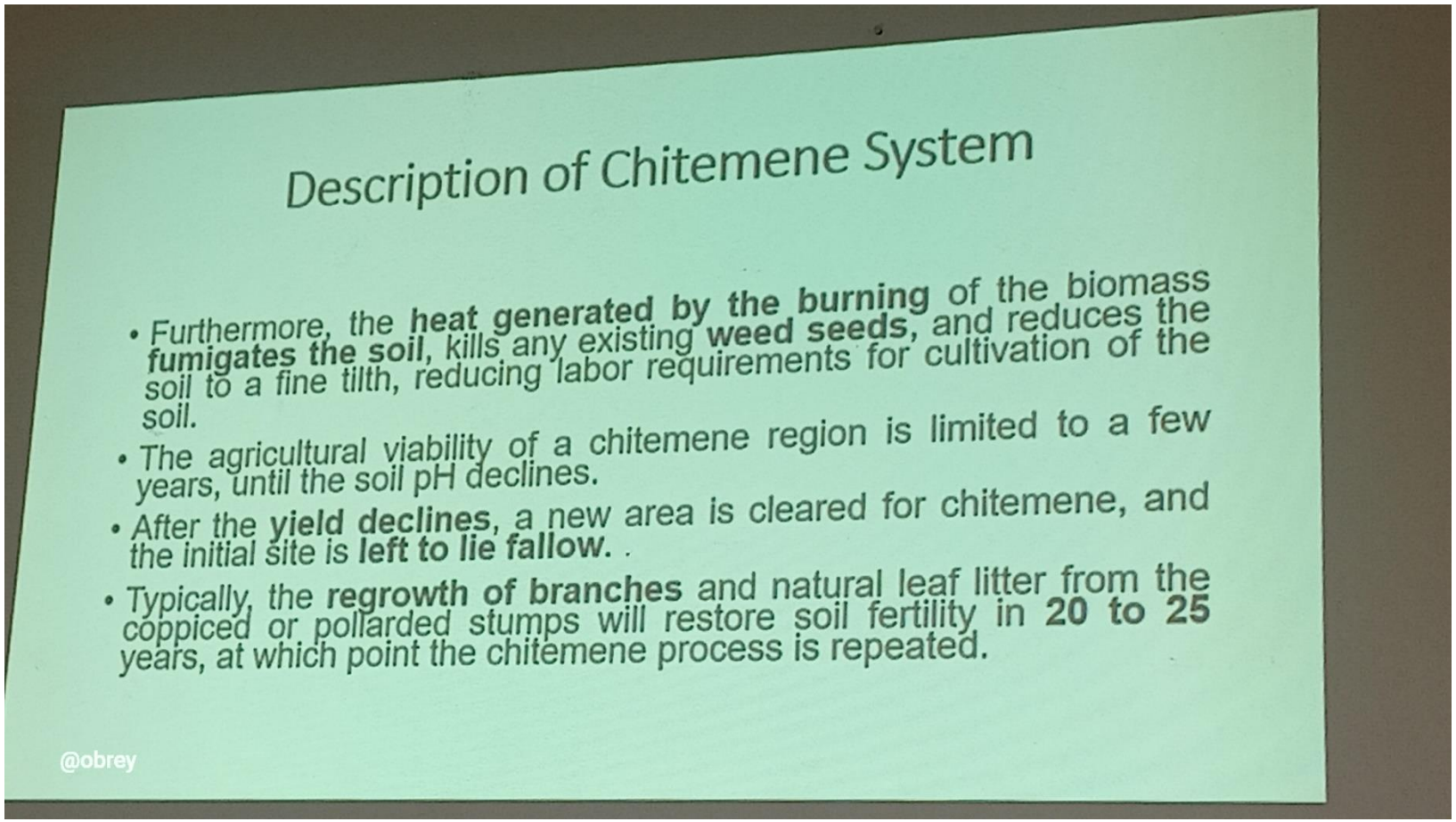

Why is chitemene system commonly Practised in Zambia?
• This kind of traditional farming is what is well know as cut and slash method.
• It is mostly practiced in the northern part of Zambia.
• People use the system very much because it is the cheap way and method to neutralise acidic soil for growing crops such as finger millet.
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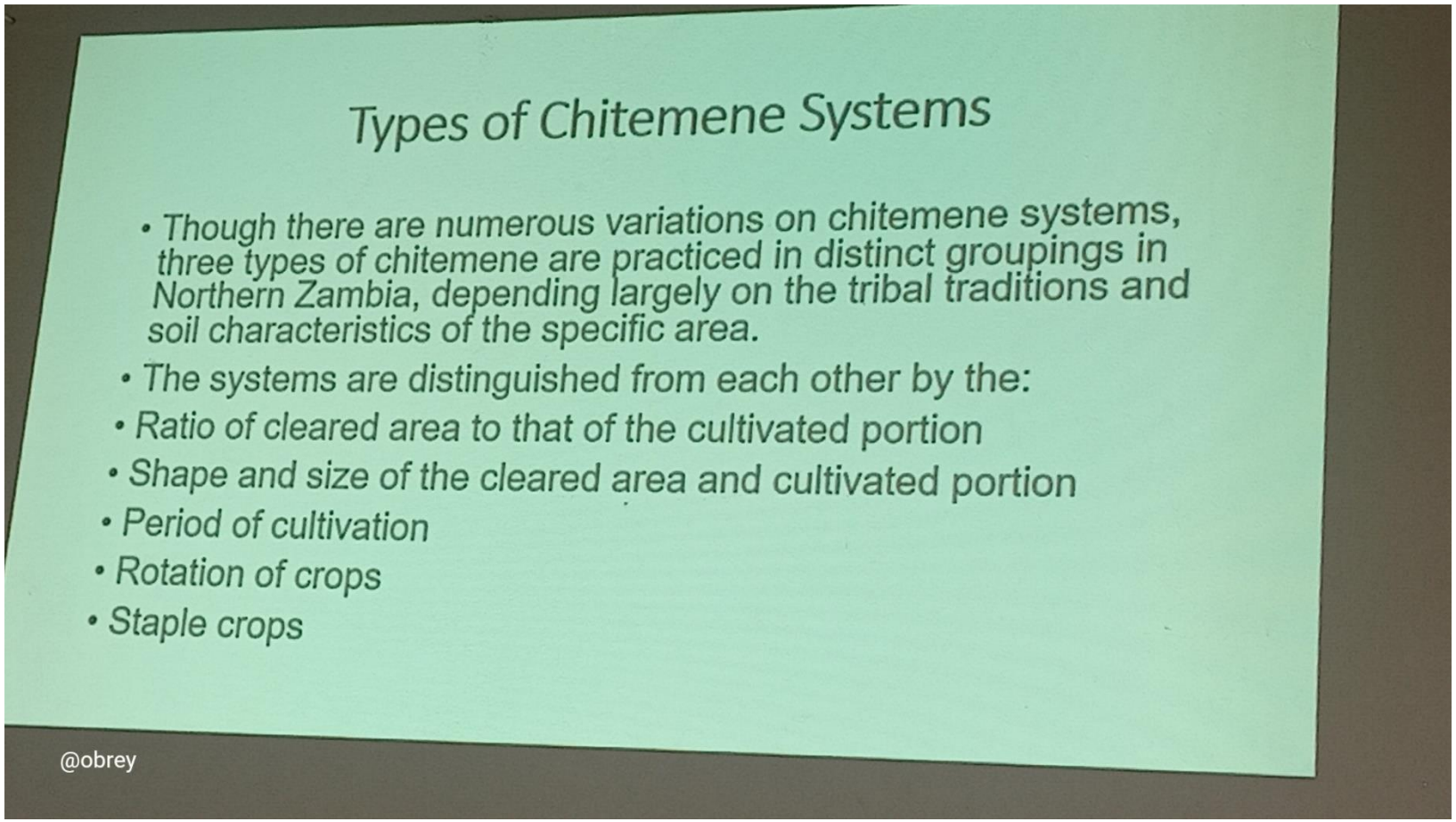
The main features of the chitemene system?

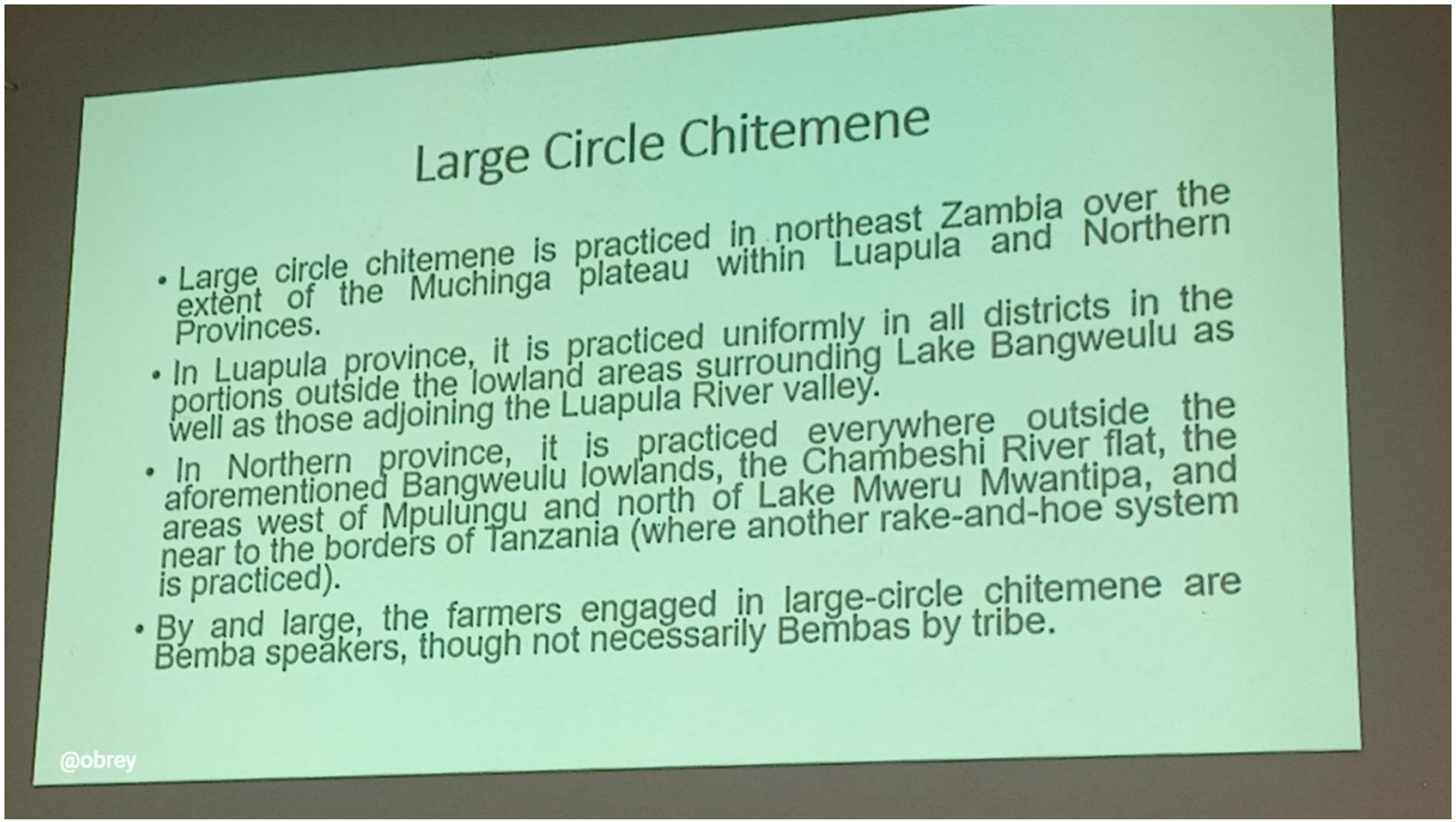
- Main features include:
- **Coppicing of trees** and **stacking of trunks** and branches into a series of long, rectangular-shaped mounds.
- Unlike large-circle chitemene, the cleared area may contain more than one mound to be burned or cultivated.
- Immediate cultivated land use is for **two to three years**

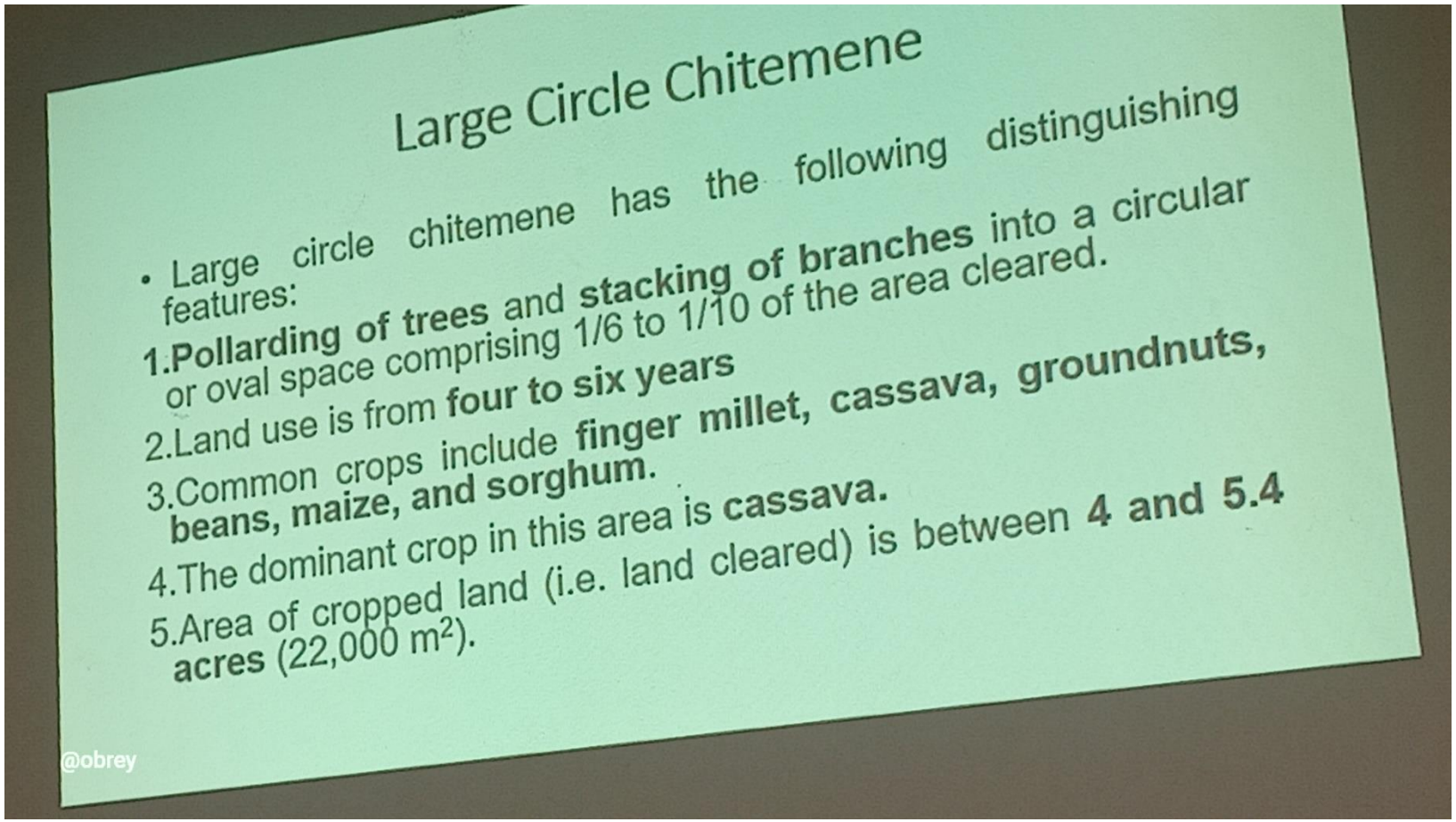
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Description of Chitemene System
• Chitemene systems are most widely used throughout the Central Zambezian Miombo woodlands that is the largest ecoregion in Zambia
• It includes Northwestern, Copperbelt, Central, Northern, and Luapula provinces.
• Typical soils in this biome are of the **order oxisols,** which are highly weathered, acidic, and easily leached soils.
• The pH of these soils range **from 4.0 to 4.5**, values too acidic for the cultivation of most common **cereal grains** and **root crops** in Zambia (maize, finger millet, sorghum, and cassava).
• The chitemene system, which creates a **surplus of ash** in concentrated spaces, **raises the soil pH**, enabling the cultivation of those crops.
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Description of Chitemene System
• Furthermore, the heat generated by the burning of the biomass fumigates the soil, kills any existing weed seeds, and reduces the soil to a fine tilth, reducing labor requirements for cultivation of the soil.
• The agricultural viability of a chitemene region is limited to a few years, until the soil pH declines.
• After the yield declines, a new area is cleared for chitemene, and the initial site is left to lie fallow. .
• Typically, the regrowth of branches and natural leaf litter from the coppiced or pollarded stumps will restore soil fertility in 20 to 25 years, at which point the chitemene process is repeated.
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Types of Chitemene Systems
• Though there are numerous variations on chitemene systems, three types of chitemene are practiced in distinct groupings in Northern Zambia, depending largely on the tribal traditions and soil characteristics of the specific area.
• The systems are distinguished from each other by the:
• Ratio of cleared area to that of the cultivated portion
• Shape and size of the cleared area and cultivated portion
• Period of cultivation
• Rotation of crops
• Staple crops
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Large Circle Chitemene
• Large circle chitemene is practiced in northeast Zambia over the extent of the Muchinga plateau within Luapula and Northern Provinces.
• In Luapula province, it is practiced uniformly in all districts in the portions outside the lowland areas surrounding Lake Bangweulu as well as those adjoining the Luapula River valley.
• In Northern province, it is practiced everywhere outside the aforementioned Bangweulu lowlands, the Chambeshi River flat, the areas west of Mpulungu and north of Lake Mweru Mwantipa, and near to the borders of Tanzania (where another rake-and-hoe system is practiced).
• By and large, the farmers engaged in large-circle chitemene are Bemba speakers, though not necessarily Bembas by tribe.
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Large Circle Chitemene
• Large circle chitemene has the following distinguishing features:
1.Pollarding of trees and stacking of branches into a circular or oval space comprising 1/6 to 1/10 of the area cleared.
2.Land use is from four to six years
3.Common crops include finger millet, cassava, groundnuts, beans, maize, and sorghum.
4.The dominant crop in this area is cassava.
5.Area of cropped land (i.e. land cleared) is between 4 and 5.4 acres (22,000 m²).
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Small Circle Chitemene

- This system is practiced in an area extending from the southern portion of Mpika District (Muchinga Province) westwards throughout the Muchinga Escarpment uplands of Central Province to the vicinity of Kapiri Mposhi.
- It is a common practice of the predominant tribes of the region, the Laala and Swaka peoples.
- Its main features include:

1. **Coppicing** of trees and **stacking of trunks** and branches into a series of long, rectangular-shaped mounds.
2. Unlike large-circle chitemene, the cleared area may contain **more than one mound to be burned** or cultivated.
3. Immediate cultivated land use is for **two to three years.**
4. Predominant crops are **millet, maize, and sorghum.**

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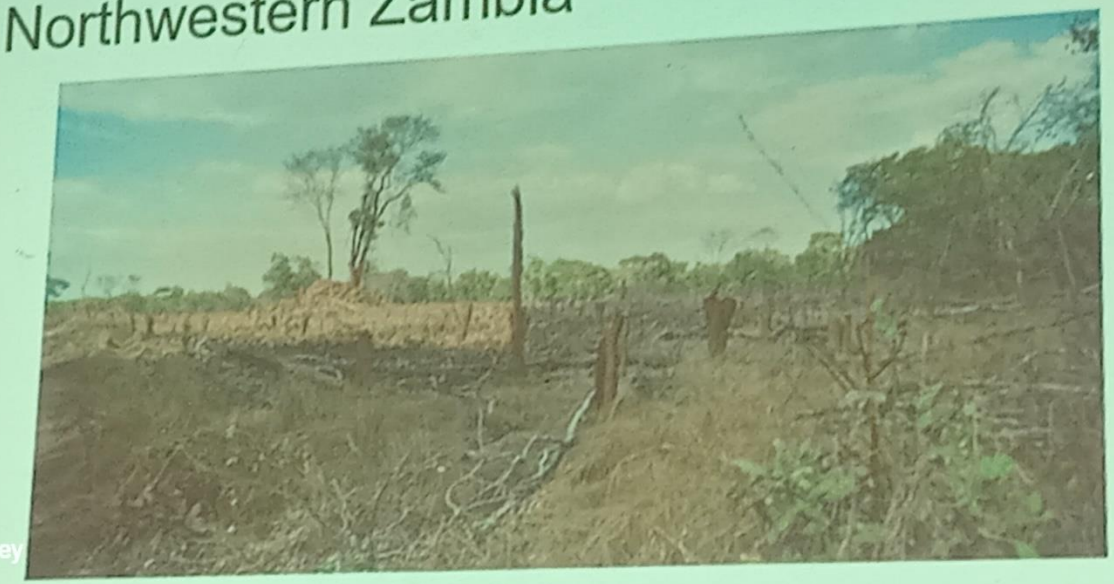
Block Chitemene System

- This system is predominant from Kapiri Mposhi in the east to Mufumbwe in the west, and is a practice of the baLamba and baKaonde tribes of Central, Copperbelt, and Northwestern Provinces.

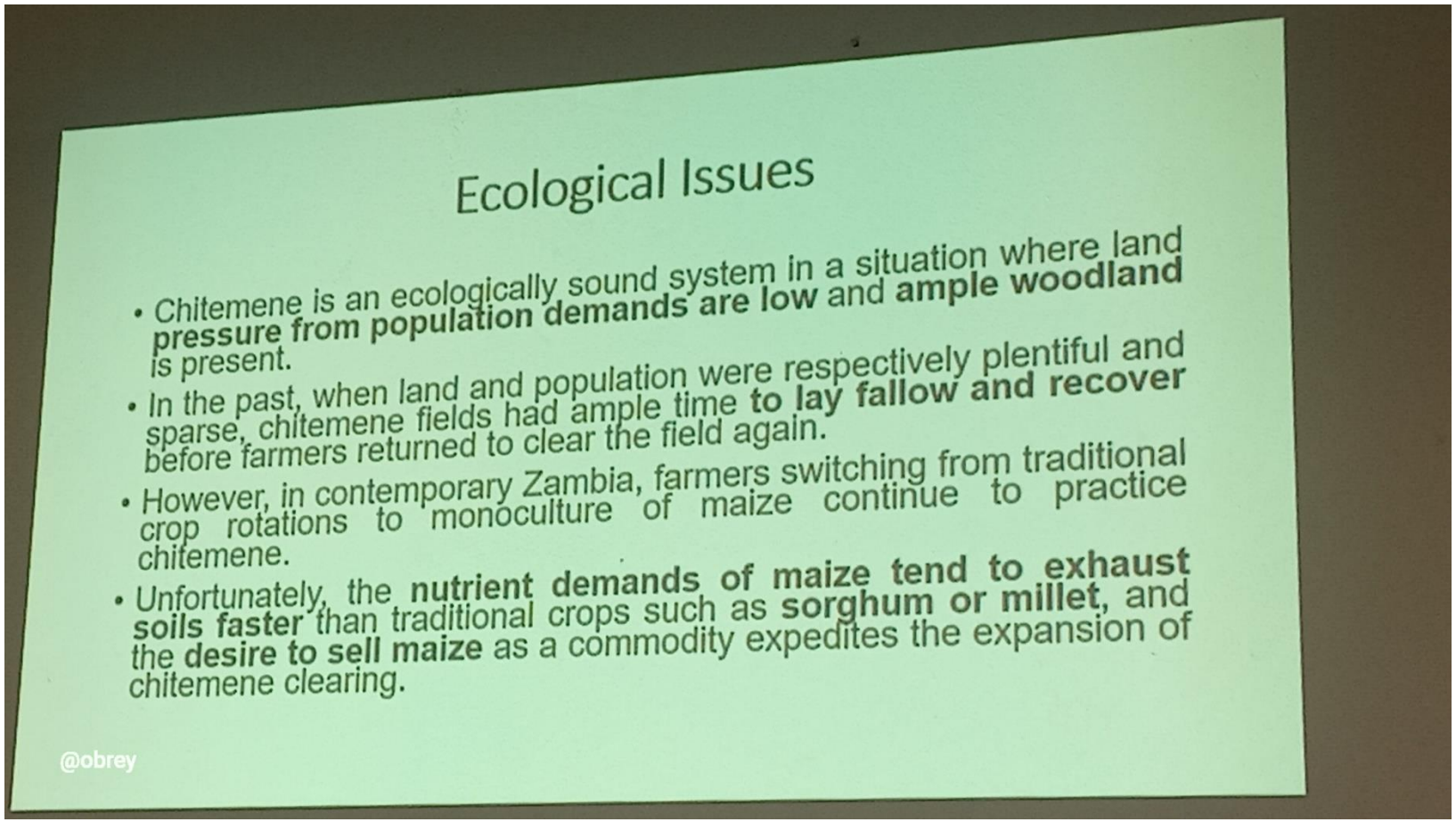
1. Due to the relative small number of people and abundance of land, **large trees are base-cut, left to dry, and then burned before the rains.**
2. Crops are planted in both the **cleared and burned areas** to the extent that nearly all cleared land is **planted.**
3. Size of cleared land is approximately **2.3 acres** (9,300 m^2).
4. Predominant crops are **maize and sorghum**

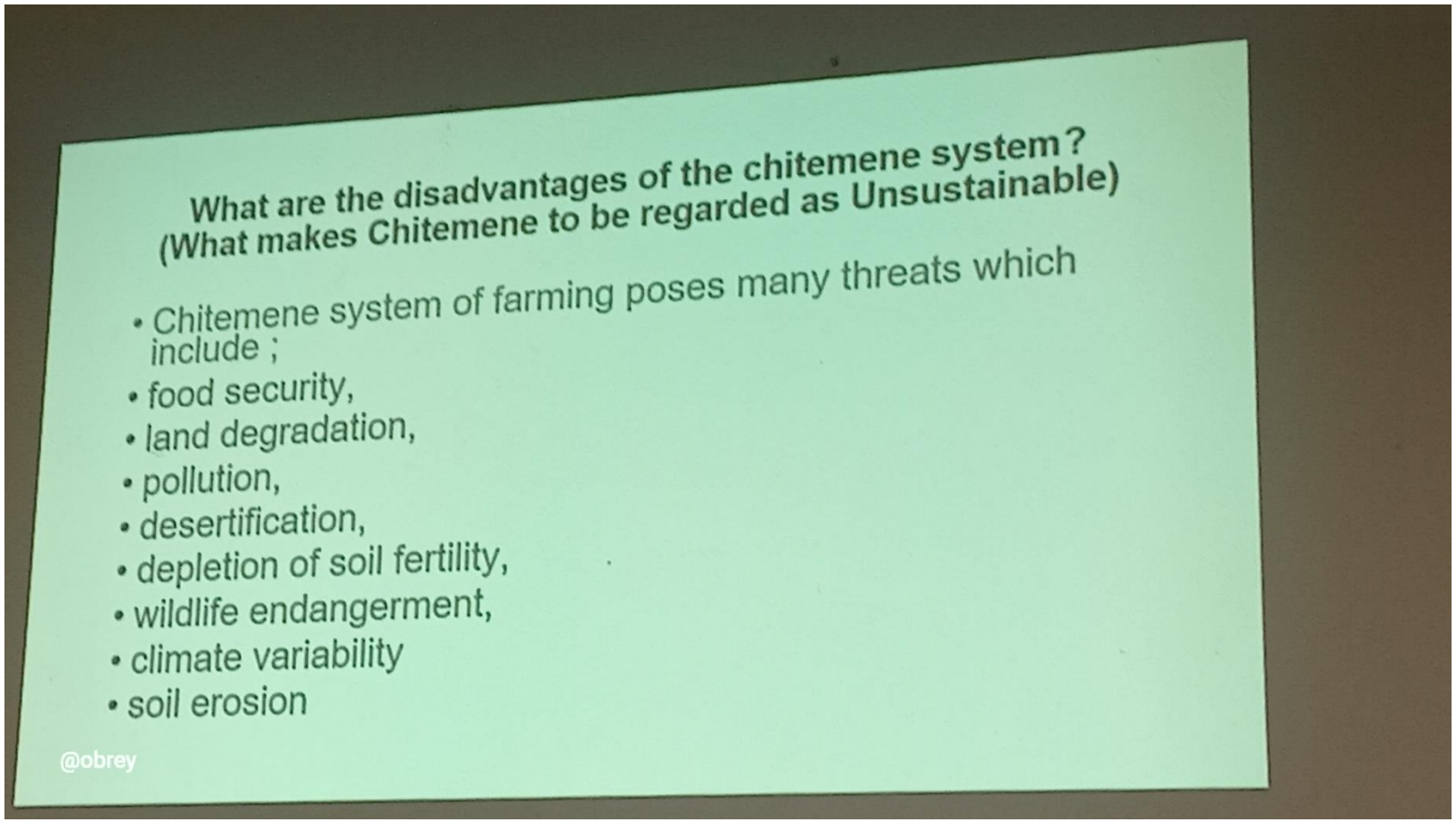
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Block Chitemene" in Kasempa District, Northwestern Zambia



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Ecological Issues
• Chitemene is an ecologically sound system in a situation where land pressure from population demands are low and ample woodland is present.
• In the past, when land and population were respectively plentiful and sparse, chitemene fields had ample time to lay fallow and recover before farmers returned to clear the field again.
• However, in contemporary Zambia, farmers switching from traditional crop rotations to monoculture of maize continue to practice chitemene.
• Unfortunately, the nutrient demands of maize tend to exhaust soils faster than traditional crops such as sorghum or millet, and the desire to sell maize as a commodity expedites the expansion of chitemene clearing.
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What are the disadvantages of the chitemene system?
(What makes Chitemene to be regarded as Unsustainable)
• Chitemene system of farming poses many threats which include ;
• food security,
• land degradation,
• pollution,
• desertification,
• depletion of soil fertility,
• wildlife endangerment,
• climate variability
• soil erosion
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